Private qualifier explicit inspection directory

Task T-1459. 3 November 2026.

Observed failure	Default temporary directories unavailable
Minimal correction	Anonymous file in owned command directory
Source controls	14 preserved + 8 targeted controls passed
Runtime state	No retry; successor admission separate

Preserve read-only system protection and existing bounded writable storage. T-1459.

The new qualifier source creates anonymous inspection files in the existing owned command directory. All 22 software controls passed: 14 preserved controls and eight focused temporary-file checks. The change is confined to a small helper and its call site. Runtime qualification remains unadmitted pending separate review and successor admission.

Failure and diagnosis. The retained v3 qualification receipt reports no passed cases and FileNotFoundError: no usable temporary directory among /tmp, /var/tmp, /usr/tmp and /. The first daemon-info inspection allocates TemporaryFile before launching its client. Supervisor-v3 uses ProtectSystem=strict with narrow writable exceptions, and its explicit callback environment supplies no TMPDIR, TEMP or TMP. Python3.9 tempfile searches those global defaults and the current directory when dir is absent. This source chain explains the retained error; no live namespace was re-entered for diagnosis.

Correction. inspection_capture(directory) calls TemporaryFile(mode='w+b',dir=directory). command passes d, which it has just created below the owned qualification receipt path on the bounded private filesystem. The explicit directory bypasses ambient temporary-directory search. File lifetime remains anonymous or immediately unlinked under Python's O_TMPFILE/fallback behavior; normal closure and exception closure leave no persistent raw inspection file. An allocation error still occurs before Popen. Existing raw hash/size recording and sanitized inspection receipts stay unchanged.

Controls. A forced unavailable default resolver reproduces the old failure and the new helper succeeds without consulting it. A poisoned TMPDIR/TEMP/TMP environment is ignored. Real local-file tests check binary roundtrip, regular-file type, mode 0600, zero links and closure; a forced non-O_TMPFILE path checks the unlinked fallback. Missing directory and mocked EROFS are rejected without global fallback. An exception closes the descriptor. The final ordering assertion is static AST inspection, not a live process-launch test. None of these controls reproduces a systemd mount namespace.

Verification. The single prospective wrapper was limited to 120 seconds and 2 GiB address space; it completed in 0.203562 seconds, returned exit 0, reaped its child and retained unchanged source/test hashes. unittest reports 22 tests in 0.029 seconds. The preserved controls cover endpoint ownership, deadlines, release-file custody, gate ordering, resource flags and strict diff. Gate source, private contract, HEAD contract and draft binding template are byte-identical to v2. Both admission booleans remain false. No container, Docker daemon, systemd service, mount or cloud operation was invoked.

Boundary. The correction resolves the source dependency on a writable global temporary directory. Independent review and a separately authorized successor qualification are needed to check the complete live path. Previous source packets, failed runtime receipt, gate binary and mathematical evidence are preserved. No global writable paths or service-hardening changes are proposed.